FINC5001

Module 6

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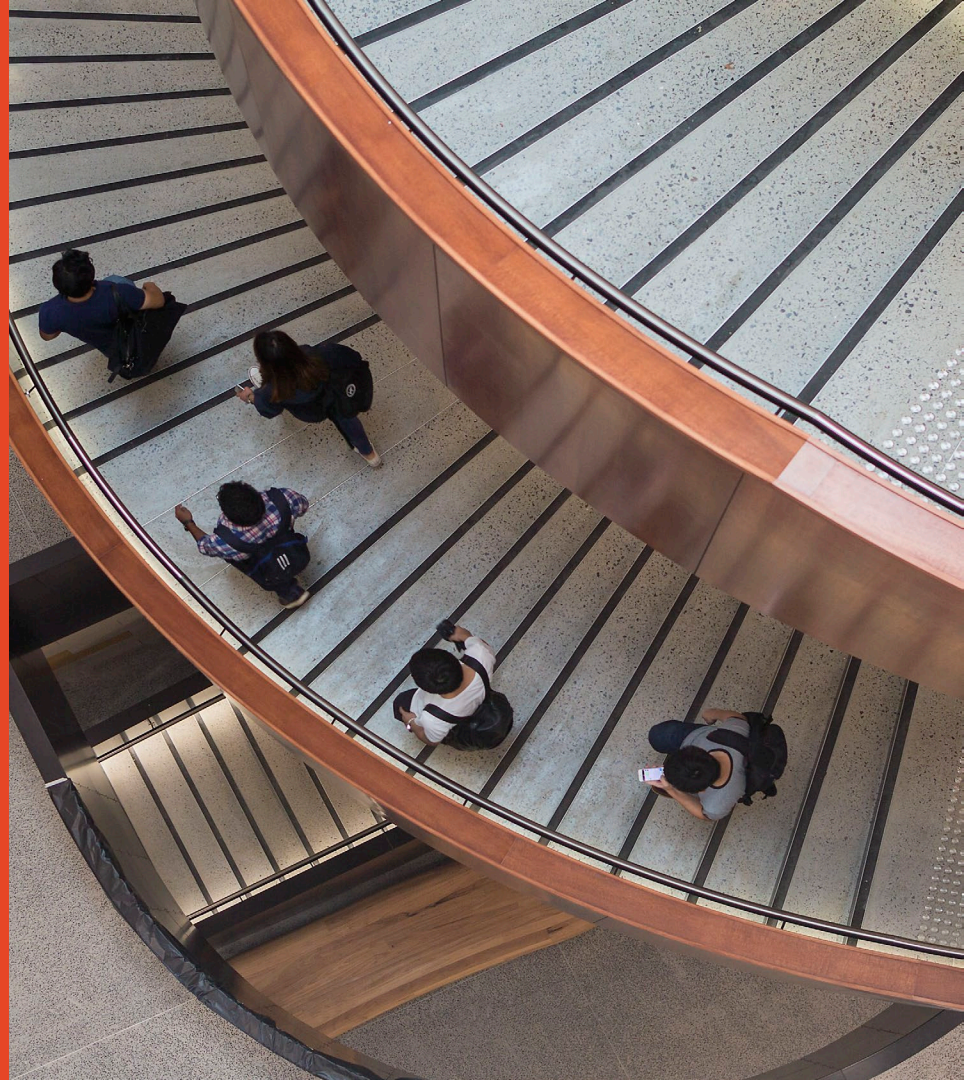


Overview

- In this lecture we will:
  - Recap the components of total risk
  - Derive the CAPM equation
  - Examine two major CAPM issues:
    - Estimation of the parameters of the CAPM
    - Adjusting beta for leverage
  - Portfolio performance evaluation
    - Sharpe ratio
    - Taynor Ratio
    - Jensens Alpha
  - Arbitrage Pricing Model (APT)
  - Three Factor Model

Recap: Components of risk

- The risk of investing in a single security can be reduced by combining the security with others in a portfolio
- **However, this does not mean that all risk can be eliminated**
- This is because there are two components of total risk
  - Unsystematic (or diversifiable) risk
  - Systematic (or non-diversifiable) risk

Recap: Unsystematic risk

- Unsystematic risk relates to price movements that are caused by an event that influences a single company alone
- It is known as diversifiable risk because it can be diversified away
- *Example:* The BHP price fell 4% on the morning of 19 September 1997, even though the All Ordinaries was relatively stable. This was caused by a poor quarterly earnings announcement by BHP

Recap: Systematic risk

- Systematic risk relates to macroeconomic events that affect the prices of all securities, and are reflected in broad market movements
- It is referred to as 'non-diversifiable' as it is common to all securities and cannot be diversified away
- *Example:* Price movement caused by the Asian currency crisis of October 1997 had a dramatic effect on the Australian market

A well-diversified portfolio faces market risk alone, and the level of market risk will depend on the market risk of stocks included in the portfolio

Measurement of the components of total risk

- **How can we measure market risk?**
- Measure how sensitive movements in the stock are to movements in the market
- Substituting into the equation for a straight line we can obtain:

$$r_i = \alpha + \beta r_m$$

where:

r_m = return on the market

r_i = return on stock i

α = the intercept coefficient (Y intercept)

β = the slope coefficient (slope of the line)

- This is known as the ***market model***

Beta

- The formula for calculating Beta becomes:

$$\beta_i = \frac{Cov(R_m, R_i)}{\sigma_m^2}$$

- β_i = Beta for stock i
- R_m = Returns for the market
- R_i = Returns for stock i
- σ_m^2 = Variance for the market

Measurement of the components of total risk (cont)

– Beta (β) measures the degree of exposure of the individual stock *i* to market risk

IF	INTERPRETATION
$\beta = 1$	Share as risky as market
$\beta > 1$	Share more risky than market
$\beta < 1$	Share less risky than market

Portfolio Beta

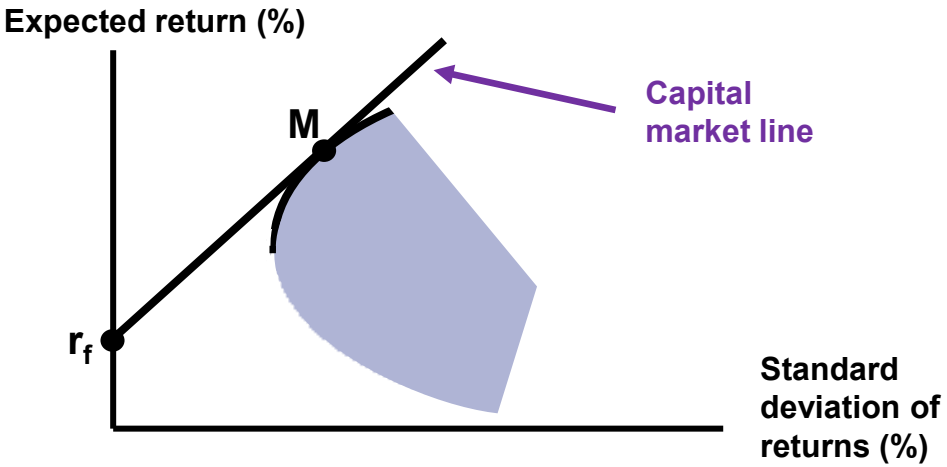
- Portfolio beta measures the systematic risk of a portfolio of stocks
- Portfolio beta (n stocks):

$$\beta_p = \omega_1\beta_1 + \omega_2\beta_2 + \cdots + \omega_n\beta_n$$

- β_p = Portfolio beta
- ω_i = Weight on stock i
- β_i = Beta of stock i

- **The portfolio beta is the weighted average of all the individual betas.**

Recap: optimum portfolios – multiple stocks plus a risk-free asset (cont)



- The line is called the *capital market line* (CML), and the point of tangency is represented by the portfolio M
- M is a theoretical portfolio containing all available stocks
- It is the *market portfolio*, and hence its level of risk and return are equal to those of the market overall

Recap: CML

- The equation for the CML gives us an expression for the expected return on any efficient portfolio (i.e. a portfolio that lies on the CML):

$$r_p = r_f + \frac{r_m - r_f}{\sigma_m}\sigma_p$$

where:

r_p, r_f, r_m = return on the portfolio, the risk-free asset and the market, respectively

σ_p, σ_m = standard deviation of the portfolio and the market, respectively

Recap: CML

- This equation can be rearranged to give

$$r_p = r_f + \left(r_m - r_f\right)\frac{\sigma_p}{\sigma_m}$$

- This is an ***equilibrium pricing relationship*** – it describes the return expected by investors from holding a portfolio of stocks

The CAPM

- The expected ratio of the standard deviation of the portfolio to that of the market (σ_p/σ_m) is the beta (β) of the portfolio:

$$E(r_p) = r_f + E(r_m - r_f)\beta_p$$

where:

E(•) denotes an expected outcome

- To describe how *individual securities* are priced:

$$E(r_i) = r_f + E(r_m - r_f) \times \beta_i$$

where:

r_f	=	the risk-free rate of return
$E(r_i)$	=	expected return on security *i*
$E(r_m)$	=	expected return on the market
β_i	=	beta of security *i*

The CAPM

- The last equation is the CAPM
- This equation can be used to determine the expected return on any security
- It is consistent with the notion that the price (or expected return) of a security is determined by its market risk
- This is because rational investors will hold a well-diversified portfolio, and the risk of a well-diversified portfolio is market risk

Another way of thinking about the CAPM

- Return on a security is the sum of the risk-free rate and a risk premium:

$$E(r_i) = r_f + \text{ risk premium}$$

- Since investors hold assets in portfolios (market portfolios) the only portion of risk that they take into consideration in pricing assets is market risk
- The quantity of market risk attributable to an asset is given by its β

Another way of thinking about the CAPM (cont)

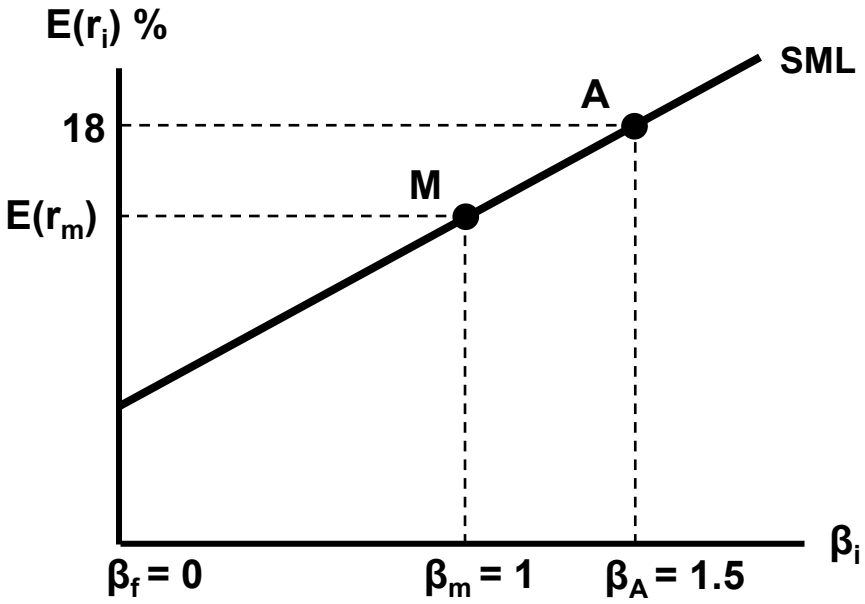
- For each unit of market risk faced, the investor expects to be remunerated by a market rate of return over and above the risk-free rate ($r_m - r_f$)
- This is the price of market risk
- Hence, the risk premium is the product of the quantity of market risk and the price of market risk faced by the investor:

$$\begin{aligned} &\text{risk premium} \\ &= \text{quantity of risk} \times \text{price of risk} \\ &= \beta_{\mathrm{p}} \times E\left(r_m - r_f\right) \end{aligned}$$

The security market line

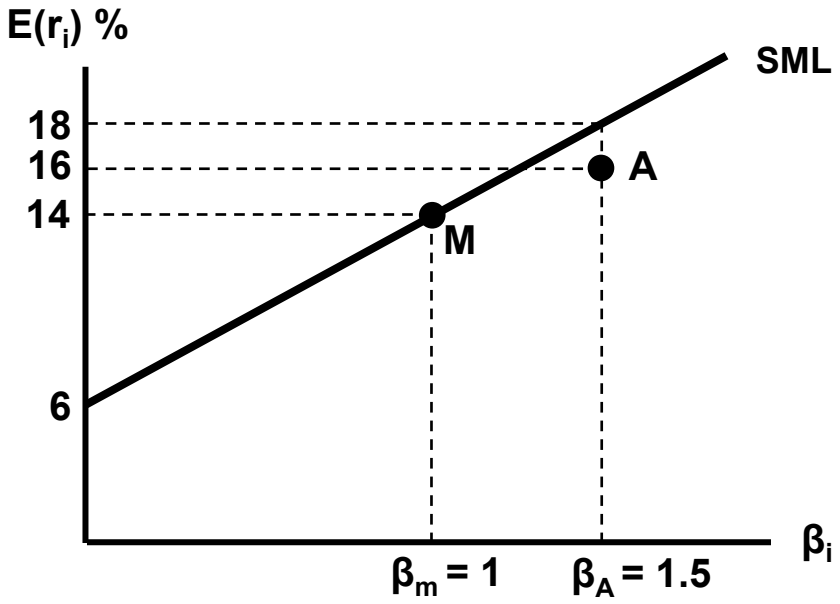
- The CAPM can be graphically represented by a relationship known as the ***security market line***
- Once the risk-free rate and the expected return on the market have been estimated, the expected return on a security can be calculated as a function of its beta and should plot on the SML
- For example, the risk-free rate is 6%, the expected market return is 14% and a stock has a β of 1.5

The security market line (cont)



– The CAPM gives an expected return of 18%, and the stock plots on the SML as shown

The security market line (cont)



– What happens if the observed return differs from that predicted by the CAPM and the SML?

The security market line (cont)

- In terms of the CAPM, this security is not priced in equilibrium – however, this disequilibrium will only be temporary
  - Security A is currently overpriced – its return is below that predicted as commensurate with market risk (beta)
  - Investors are not being fully compensated for risk and would have an incentive to sell security A
  - This selling pressure would push down the price of security A (increasing its expected return)
  - This would cease when the price of security A is reduced to the level at which return equals 18%

Estimation of beta in practice

- In order to determine the expected return on a stock, estimates are required of:
  - The risk-free rate (r_f)
  - The expected return on the market over and above the risk-free rate ($r_m - r_f$)
  - The stock's beta (β_i)
- We will examine the procedure for estimating the beta of a stock using historical data

Estimation of beta in practice (cont)

- Since β is the relationship between returns on a stock and the market, it can be estimated by quantifying this historical relationship
- The equation for the CAPM can be more generally as:

$$\boxed{Y_t = a + bX_t}$$

where:

$Y_t = E(r_{i,t} - r_{f,t})$

$X_t = E(r_{m,t} - r_{f,t})$

$r_{i,t}$ = the return on stock *i*, earned over period *t*

$r_{f,t}$ = the risk-free rate of return, earned over period *t*

$r_{m,t}$ = the market rate of return, earned over period *t*

Estimation of beta in practice (cont)

- The CAPM implies that the coefficient $a = 0$
- The coefficient b, which measures the β of stock i, is a regression coefficient that can be calculated from historical data, and is given by:

$$\beta = \frac{\text{covariance}\left(X_t, Y_t\right)}{\text{variance}\left(X_t\right)}$$

- The first step in calculating beta is to collect historical data and calculate a sequence of excess returns on a stock and on the market

Estimation of beta in practice (cont)

- The return on the stock in any month must include the dividend paid (if any) as follows:

$$r_{i,t} = \frac{P_t - P_{t-1} + d_t}{P_{t-1}}$$

where P_t is the price of the stock at the end of month t,
d_t is the dividend paid during month t (if any)

- The return on the market is simply:

$$r_{m,t} = \frac{I_t - I_{t-1}}{I_{t-1}}$$

- where I_t is the index level at the end of month t.

Estimation of beta in practice (cont)

– Once the monthly excess returns on the stock and market have been calculated by subtracting the risk-free rate from stock and market returns, the covariance of the excess returns on the stock and market can be calculated using the following expression:

$$\sigma_{im} = \frac{\sum_{t=1}^{n}\left[\left\{\left(r_{i,t} - r_{f,t}\right) - \left(\overline{r_{i,t} - r_{f,t}}\right)\right\} \times \left\{\left(r_{m,t} - r_{f,t}\right) - \left(\overline{r_{m,t} - r_{f,t}}\right)\right\}\right]}{n-1}$$

where *n* is the number of observations used to estimate beta.

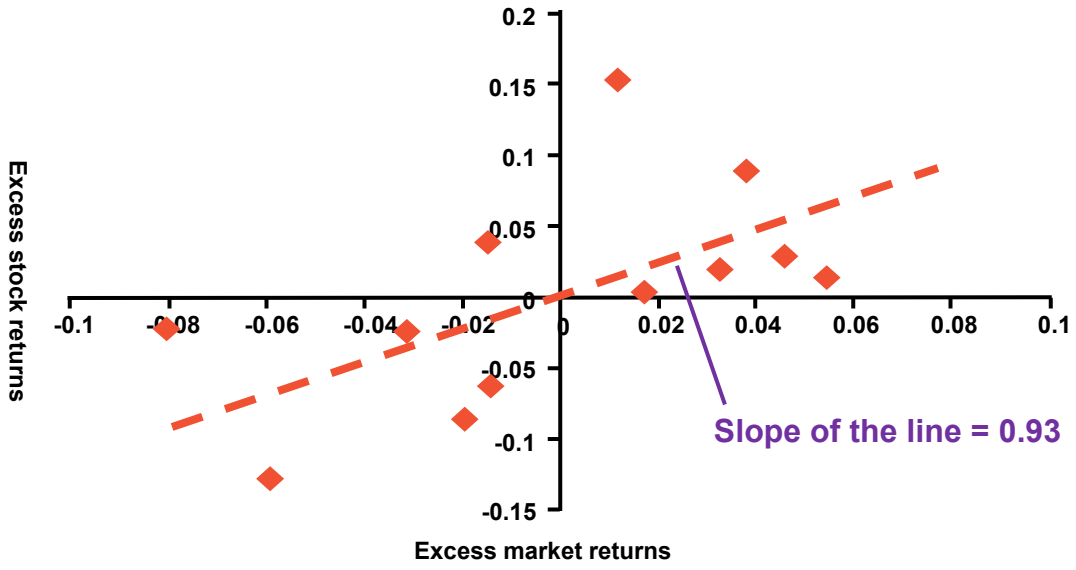
Estimation of beta in practice (cont)

- The beta is then the ratio of the covariance between excess stock returns and excess market returns to the variance of excess market returns
- This is further illustrated by considering the *characteristic line*, which is the line that best fits the relationship between excess returns on the stock and the market
- The slope of the characteristic line is the beta of the stock – as shown in the following example

Estimation of beta in practice (cont)

EXAMPLE:

Characteristic line for Wesfarmers



Adjusting β for Leverage

- Often estimation of β is problematic because the company:
  - has insufficient price history
  - has undergone a structural change
  - is considering investment in a new business
- In such cases may need to estimate β from another company, and adjust its β to reflect different capital structure

Adjusting β for alternative capital structures

- Since the debt of a company increases the cashflow risk of shareholders:

$$\beta_l = \beta_u[1+(1-T)\ (D/E)]$$

Where:

β_l = levered Beta
β_u = unlevered Beta
T = company tax rate
D/E = debt/equity ratio

CAPM in Practice

Practical Issues with CAPM:

1. Estimating beta is very difficult
2. Finding the market portfolio is difficult
3. Determining expected returns is difficult

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Alpha

- How do we know how good a fund manager is?

- **Alpha** - Alpha is a measure of the skill or performance of an investment. It is the difference between the expected return predicted by the CAPM and the actual return of the investment.

- Alpha is also called **abnormal return.**

- When related to CAPM, it is also sometimes called **Jensen's alpha**.

Alpha

- The formula for alpha is:

$$\alpha = ER - \left[r_f + \beta\left(E(R_m) - r_f\right)\right]$$

- α = Alpha
- ER = Expected return of stock (or portfolio)
- r_f= Risk-free rate
- β = Beta of the stock (or portfolio)
- ER_m = Expected return of market

Alpha

- It is important to note that ER here is not the same as the CAPM expected return but our best estimate using the *actual returns* of the investment. Thus, alpha is always calculated *after* returns have been realised.
- Alpha can also be positive, negative, or zero.
- Then a positive alpha implies that the investment is doing better than the CAPM predicts. That is, the investment is generating more expected returns while taking less risk.
- Similarly, a negative alpha implies that the investment is doing worse than the CAPM predicts.

Performance Evaluation

- A simple manner to evaluate the performance of an investment is to consider its arithmetic and geometric average rate of return.

- The average return can be compared to a benchmark, such as an appropriate market index or the median return of funds in a comparison group.

- **Risk-adjusted** performance evaluation methods were proposed soon after the Capital Asset Pricing Model (CAPM) theory was developed.

Performance Evaluation

$$Sharpe\ Ratio = \frac{E(R_p) - r_f}{\sigma_p}$$

- $E(R_p)$ = Expected return of portfolio
- r_f = Risk-free rate
- σ_p = Standard deviation of portfolio

Performance Evaluation

$$Treynor\ Ratio = \frac{E(R_p) - r_f}{\beta_p}$$

- β_p = Beta of portfolio

Performance Evaluation

$$Jensen's\ alpha = ER - \left[r_f + \beta\left(E(R_m) - r_f\right)\right]$$

- ER = Average return of portfolio
- r_f= Risk−free rate
- β = Beta of the portfolio
- $E(R_m)$ = Expected return of market

Performance Evaluation

	Definition	Application
Sharpe ratio	$\frac{Excess\ return}{Standard\ deviation}$	Choosing among portfolios competing for the overall risky portfolio
Treynor ratio	$\frac{Excess\ return}{Beta}$	Ranking many portfolios that will be combined together to form the overall risky portfolio
Jensen's alpha	$Actual\ average\ return - return\ predicted\ by\ CAPM$	Measuring performance compared to the market portfolio

Which measure is appropriate?

Arbitrage Pricing Theory (APT)

- **Arbitrage Pricing Theory (APT)** - APT is a model that relates a stock's return to many different influences or "factors".

- APT makes different assumptions than the CAPM.

APT

- CAPM is derived through:
- Investor preferences
- Portfolio theory
- Equilibrium condition: Equilibrium means that supply equals demand

- APT:
- Assumes factors matter
- No-arbitrage condition: Investors cannot make arbitrage profits.

APT

- Factors are anything that may influence an asset's expected returns.

- Some examples of factors are (1) market factor, (2) size factor, (3) GDP factor, and (4) book value factor.

APT

For 2 factors, the APT model is:

$$ER = r_f + \beta_1(r_1) + \beta_2(r_2)$$

ER = Expected return of the stock

r_f = Risk-free rate

β_1 = Beta for factor 1

β_2 = Beta for factor 2

r_1 = Risk premium for factor 1

r_2 = Risk premium for factor 2

APT

- APT is not necessarily better or worse than CAPM. Like any model, it has its advantages and disadvantages.
- **Advantages:**

1. Does not require equilibrium
2. Allows multiple sources of systematic risk

- **Disadvantages:**

1. No theoretical justification for factors
2. Having a linear relationship is restrictive

Fama and French 3-Factor Model

$$r = R_f + \beta(R_m - R_f) + b_s \cdot SMB + b_v \cdot HML + \alpha$$

- r is the portfolio's expected rate of return
- R_f is the risk-free return rate
- R_m is the return of the market portfolio
- The "three factor" β is analogous to the classical β but not equal to it, since there are now two additional factors to do some of the work
- *SMB* stands for "**S**mall [market capitalization] **M**inus **B**ig"
- *HML* for "**H**igh [book-to-market ratio] **M**inus **L**ow"
- alpha is the error term

The End

Next week we will look at a model for determining the required return for the company as a whole